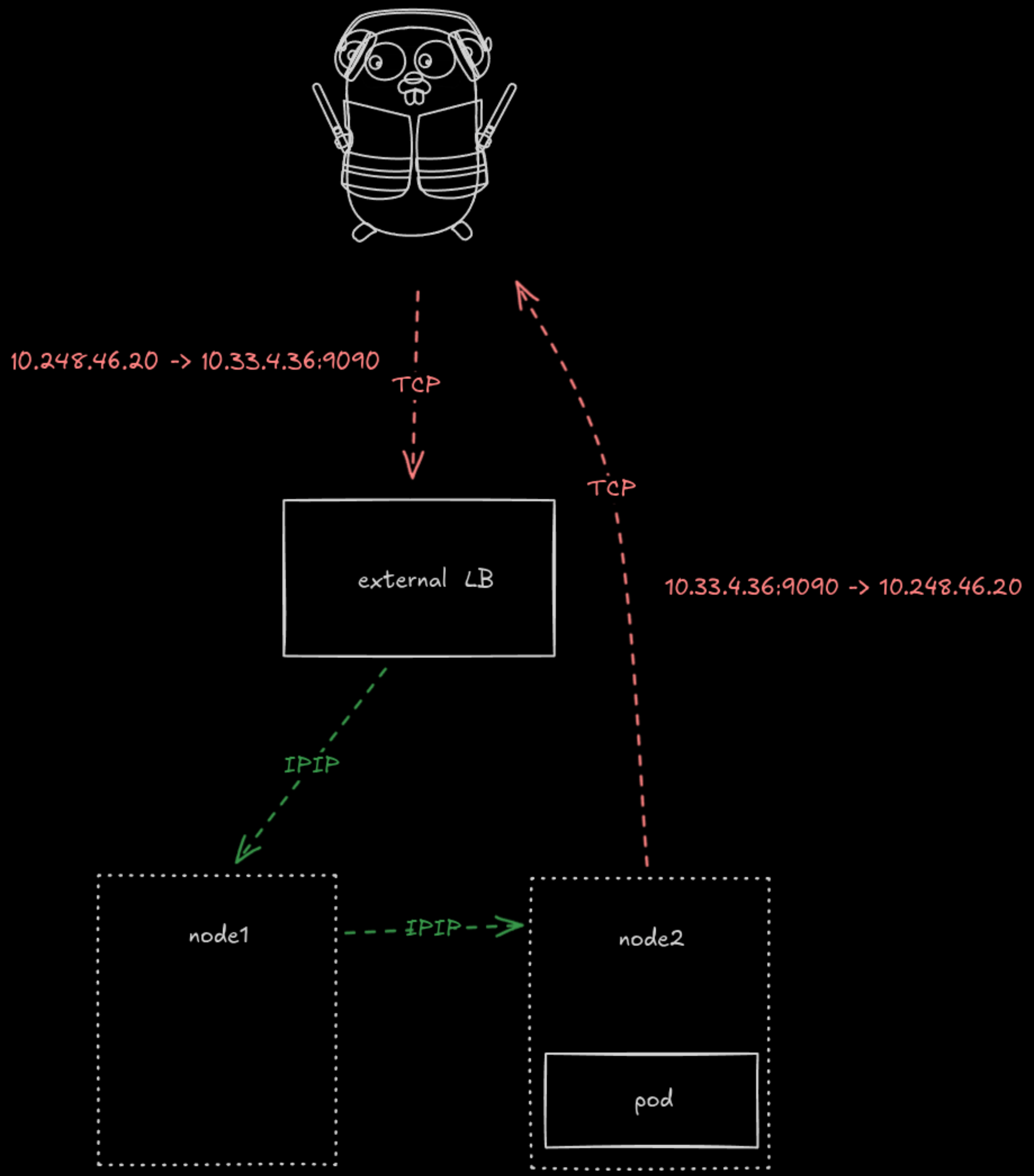
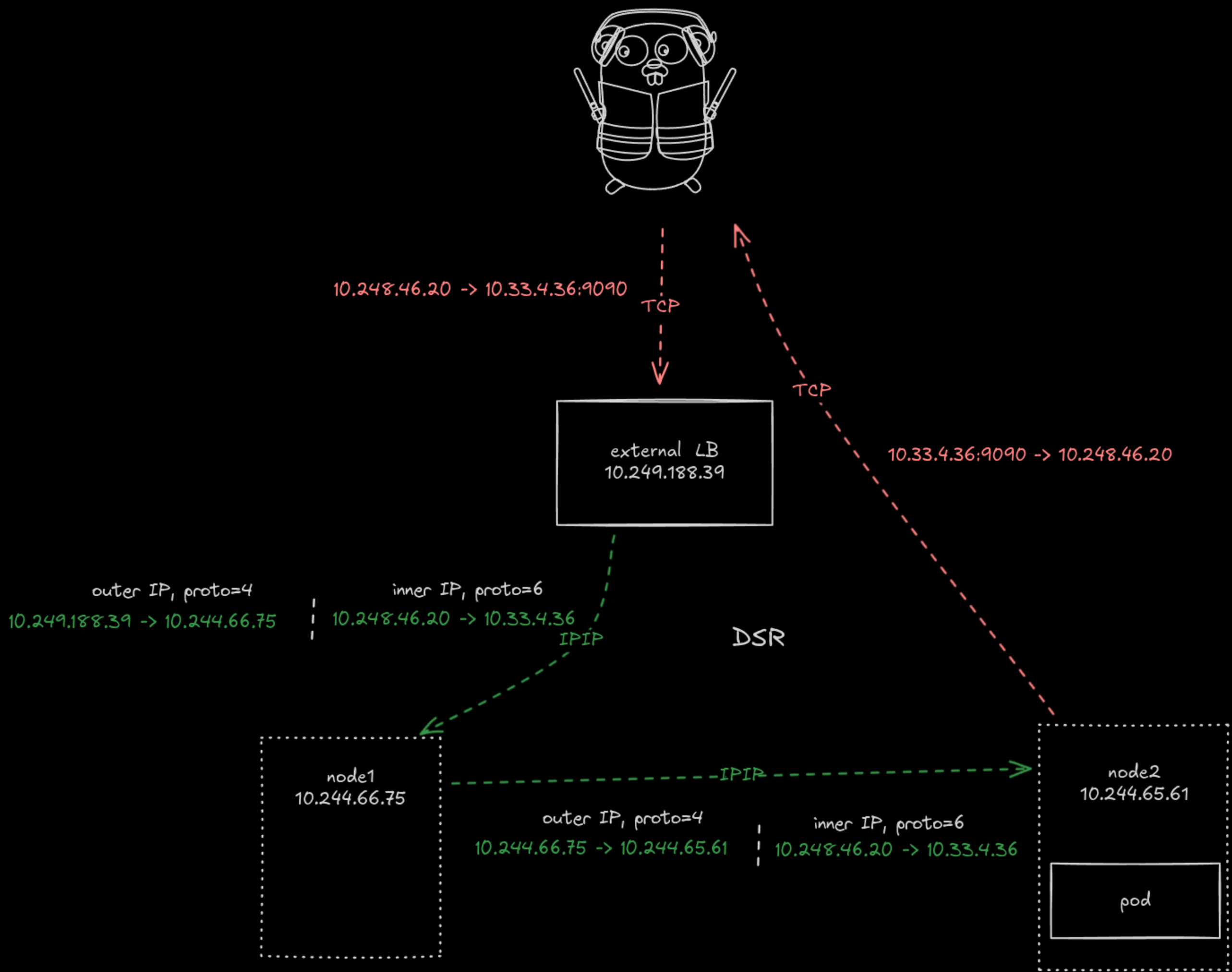
DSR/ipip encap/decap

CiliumDevSummit EU2026

@oblazek


10.248.46.20 -> 10.33.4.36:9090
TCP
TCP
external LB
10.33.4.36:9090 -> 10.248.46.20
IPIP
node1
IPIP
node2
pod


10.248.46.20 -> 10.33.4.36:9090
TCP
TCP
external LB
10.249.188.39
10.33.4.36:9090 -> 10.248.46.20
outer IP, proto=4
10.249.188.39 -> 10.244.66.75
inner IP, proto=6
10.248.46.20 -> 10.33.4.36
IPIP
DSR
node1
10.244.66.75
IPIP
outer IP, proto=4
10.244.66.75 -> 10.244.65.61
inner IP, proto=6
10.248.46.20 -> 10.33.4.36
node2
10.244.65.61
pod

```
--bpf-lb-acceleration: native
--bpf-lb-dsr-dispatch: ipip
--bpf-lb-mode: dsr
--enable-endpoint-routes=true
--enable-ipip-temination=true
```

cilium v1.13 -> v1.19+

pod-remote-node

```
enp3s0 In  IP 10.249.75.35 > 10.244.65.61: IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
cilium_ipip4 In  IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
cilium_ipip4 Out IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
enp3s0 Out IP 10.244.65.61 > 10.89.189.129: IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
```

pod-local-node

```
enp3s0 In  IP 10.244.65.61 > 10.89.189.129: IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
lxc0b70cf05bb7a Out IP 10.244.65.61 > 10.89.189.129: IP 10.248.44.163.58550 > 10.33.4.143.9090: Flags [S]
```

xx drop (CT: Unknown L4 protocol) flow 0x0 to endpoint 731, ifindex 22, file bpf_lxc.c:447, , identity world-ipv4->4404549: 10.248.46.23:39106 -> 10.33.4.143:9090 tcp SYN

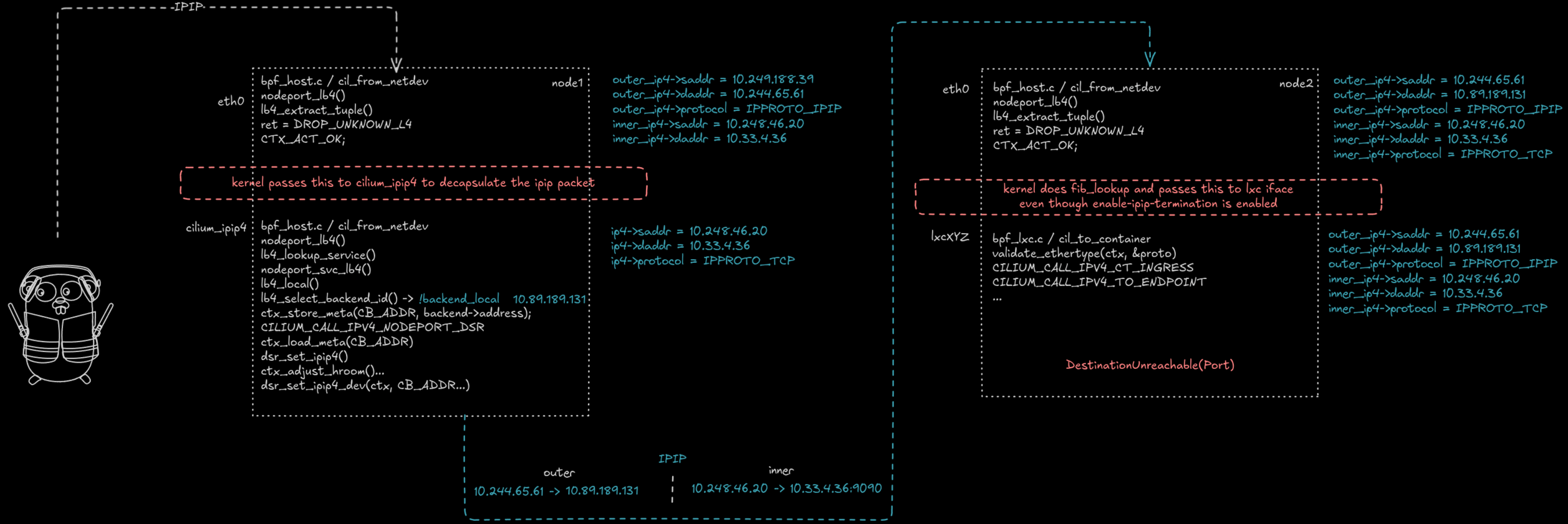
--enable-extended-ip-protocols=true

```
-> stack flow 0x0 , identity 4443831->33554633 state related ifindex 0 orig-ip 0.0.0.0: 10.89.189.129 -> 10.244.65.61 icmp DestinationUnreachable(Port)
-> network flow 0x0 , identity 4443831->unknown state unknown ifindex enp3s0 orig-ip 0.0.0.0: 10.89.189.129 -> 10.244.65.61 icmp DestinationUnreachable(Port)
```

```
enp3s0 In  IP 10.89.189.129 > 10.244.65.61: ICMP 10.89.189.129  protocol 4  port 60 unreachable, length 88
```

(on remote node)

Code
time


IPIP
node1
eth0
bpf_host.c / cil_from_netdev
nodeport_lb4()
lb4_extract_tuple()
ret = DROP_UNKNOWN_L4
CTX_ACT_OK;
outer_ip4->saddr = 10.249.188.39
outer_ip4->daddr = 10.244.65.61
outer_ip4->protocol = IPPROTO_IPIP
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
kernel passes this to cilium_ipip4 to decapsulate the ipip packet
cilium_ipip4
bpf_host.c / cil_from_netdev
nodeport_lb4()
lb4_lookup_service()
nodeport_svc_lb4()
lb4_local()
lb4_select_backend_id() -> !backend_local 10.89.189.131
ctx_store_meta(CB_ADDR, backend->address);
CILIUM_CALL_IPV4_NODEPORT_DSR
ctx_load_meta(CB_ADDR)
dsr_set_ipip4()
ctx_adjust_hroom()...
dsr_set_ipip4_dev(ctx, CB_ADDR...)
ip4->saddr = 10.248.46.20
ip4->daddr = 10.33.4.36
ip4->protocol = IPPROTO_TCP
IPIP
outer
10.244.65.61 -> 10.89.189.131
inner
10.248.46.20 -> 10.33.4.36:9090
node2
eth0
bpf_host.c / cil_from_netdev
nodeport_lb4()
lb4_extract_tuple()
ret = DROP_UNKNOWN_L4
CTX_ACT_OK;
outer_ip4->saddr = 10.244.65.61
outer_ip4->daddr = 10.89.189.131
outer_ip4->protocol = IPPROTO_IPIP
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
inner_ip4->protocol = IPPROTO_TCP
kernel does fib_lookup and passes this to lxc iface
even though enable-ipip-termination is enabled
lxcXYZ
bpf_lxc.c / cil_to_container
validate_ethertype(ctx, &proto)
CILIUM_CALL_IPV4_CT_INGRESS
CILIUM_CALL_IPV4_TO_ENDPOINT
...
DestinationUnreachable(Port)
outer_ip4->saddr = 10.244.65.61
outer_ip4->daddr = 10.89.189.131
outer_ip4->protocol = IPPROTO_IPIP
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
inner_ip4->protocol = IPPROTO_TCP

XDP

eth0

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_extract_tuple()
ret = DROP_UNKNOWN_L4
```

```
ip4->saddr = 10.249.188.39
ip4->daddr = 10.244.65.61
ip4->protocol = IPPROTO_IPIP
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
```

kernel passes this to cilium_ipip4 to decapsulate the ipip packet

SKB

cilium_ipip4

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_lookup_service()
nodeport_svc_lb4()
lb4_local()
lb4_select_backend_id() -> !backend_local
ctx_store_meta(CB_ADDR, backend->address);
CILIUM_CALL_IPV4_NODEPORT_DSR
backend_addr = ctx_load_meta(CB_ADDR)
dsr_set_ipip4()
+host_ip = lookup_ip4_remote_endpoint(backend_addr);
dsr_set_ipip4_dev(ctx, host_ip...)
ctx_set_tunnel_key(ctx, &bpf_tunnel_key...)
CTX_ACT_REDIRECT;
```

```
ip4->saddr = 10.248.46.20
ip4->daddr = 10.33.4.36
ip4->protocol = IPPROTO_TCP
```

node1
10.244.65.61

kernel passes this to cilium_ipip4 to encapsulate the ipip packet

IPIP

outer
10.244.65.61 -> 10.244.66.75

inner
10.248.46.20 -> 10.33.4.36:9090

XDP

eth0

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_extract_tuple()
ret = DROP_UNKNOWN_L4
```

```
ip4->saddr = 10.244.65.61
ip4->daddr = 10.244.66.75
ip4->protocol = IPPROTO_IPIP
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
```

node2
10.244.66.75

kernel passes this to cilium_ipip4 to decapsulate the ipip packet

SKB

cilium_ipip4

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_lookup_service()
nodeport_svc_lb4()
lb4_local()
lb4_select_backend_id() -> backend_local
lb4_dnat_request()
CTX_ACT_OK;
```

```
ip4->saddr = 10.248.46.20
ip4->daddr = 10.33.4.36
ip4->protocol = IPPROTO_TCP
```

towards bpf_lxc

backend
10.89.189.131:80

before

pod-remote-node

```
enp3s0 In  IP 10.249.75.35 > 10.244.65.61: IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
cilium_ipip4 In  IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
cilium_ipip4 Out IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
enp3s0 Out IP 10.244.65.61 > 10.89.189.129: IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
```

pod-local-node

```
enp3s0 In  IP 10.244.65.61 > 10.89.189.129: IP 10.248.46.23.39106 > 10.33.4.143.9090: Flags [S]
lxc0b70cf05bb7a Out IP 10.244.65.61 > 10.89.189.129: IP 10.248.44.163.58550 > 10.33.4.143.9090: Flags [S]
```

after

pod-remote-node

```
enp3s0 In  IP 10.249.75.35 > 10.244.65.61: IP 10.248.44.163.43348 > 10.33.4.143.9090: Flags [S]
cilium_ipip4 In  IP 10.248.44.163.43348 > 10.33.4.143.9090: Flags [S]
cilium_ipip4 Out IP 10.248.44.163.43348 > 10.33.4.143.9090: Flags [S]
enp3s0 Out IP 10.244.65.61 > 10.244.66.75: IP 10.248.44.163.43348 > 10.33.4.143.9090: Flags [S]
```

pod-local-node

```
enp3s0 In  IP 10.244.65.61 > 10.244.66.75: IP 10.248.44.163.43348 > 10.33.4.143.9090: Flags [S]
cilium_ipip4 In  IP 10.248.44.163.43348 > 10.33.4.143.9090: Flags [S]
lxc0b70cf05bb7a Out IP 10.248.44.163.43348 > 10.89.189.129.80: Flags [S]
```

```
+info = lookup_ip4_remote_endpoint(backend_addr, 0);
+if (info && info->flag_has_tunnel_ep)
+    backend_addr = info->tunnel_endpoint.ip4;

+if (dsr_set_ipip4_dev(ctx, backend_addr, 0) < 0)
+    return DROP_NO_TUNNEL_KEY;
```

```
#define lookup_ip4_remote_endpoint(addr, cluster_id) \
        ipcache_lookup4(&cilium_ipcache_v2, addr, V4_CACHE_KEY_LEN, cluster_id)
```

external LB

```
# docker exec -it cilium-l4lb cilium-dbg bpf ipcache get 10.244.65.61
10.244.65.61 maps to identity identity=9 encryptkey=0 tunnelendpoint=0.0.0.0 flags=<none>
```

k8s node

```
~ ❯ k --context infra82.ko -n cilium exec -it cilium-6nztr -c cilium-agent  -- cilium-dbg bpf ipcache get 10.89.189.129
10.89.189.129 maps to identity identity=4443831 encryptkey=0 tunnelendpoint=10.244.66.75 flags=hastunnel
```

external LB


```
{
    "metadata": {
      "name": "epslice-api",
      "namespace": "infra82-ko/default",
      "labels": {
        "kubernetes.io/service-name": "api"
      }
    },
    "addressType": "IPv4",
    "endpoints": [
      {
        "addresses": [
          "10.244.65.61"
        ],
        "conditions": {
          "ready": true,
          "serving": true,
          "terminating": false
        }
      }
    ],
    "ports": [
      {
        "name": "9090",
        "protocol": "TCP",
        "port": 9090
      }
    ]
  },
```



```
"cilium-agent",
"--bpf-lb-algorithm=maglev",
"--bpf-lb-mode=dsr",
"--bpf-lb-acceleration=native",
"--bpf-lb-dsr-dispatch=ipip",
"--routing-mode=native",
"--enable-l7-proxy=false",
"--install-iptables-rules=false",
"--enable-ipv4=true",
"--enable-k8s=false",
"--kube-proxy-replacement=true",
"--lb-state-file=/var/run/cilium/lbstate.json"
```

TBD

SKB vs XDP

XDP

eth0

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_extract_tuple()
ret = DROP_UNKNOWN_L4
...
```

HEAD =>

```
outer_ip4->saddr = 10.249.188.39
outer_ip4->daddr = 10.244.65.61
outer_ip4->protocol = IPPROTO_IPIP
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
inner_ip4->protocol = IPPROTO_TCP
```

kernel passes this to cilium_ipip4 to decapsulate the ipip packet

SKB

cilium_ipip4

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_lookup_service()
nodeport_svc_lb4()
lb4_local()
lb4_select_backend_id() -> !backend_local
ctx_store_meta(CB_ADDR, backend->address);
CILIUM_CALL_IPV4_NODEPORT_DSR
backend_addr = ctx_load_meta(CB_ADDR)
dsr_set_ipip4()
host_ip = lookup_ip4_remote_endpoint(backend_addr);
dsr_set_ipip4_dev(ctx, host_ip...)
ctx_set_tunnel_key(ctx, &bpf_tunnel_key...)
CTX_ACT_REDIRECT;
```

HEAD =>

```
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
inner_ip4->protocol = IPPROTO_TCP
```

node1
10.244.65.61

kernel passes this to cilium_ipip4 to encapsulate the ipip packet

xDP

eth0

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_extract_tuple()
```

HEAD =>
```
outer_ip4->saddr = 10.249.188.39
outer_ip4->daddr = 10.244.65.61
outer_ip4->protocol = IPPROTO_IPIP
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
inner_ip4->protocol = IPPROTO_TCP
```

```
lb4_lookup_service()
nodeport_svc_lb4()
lb4_local()
lb4_select_backend_id() -> !backend_local
ctx_store_meta(CB_ADDR, backend->address);
CILIUM_CALL_IPV4_NODEPORT_DSR
revalidate_data_l3_off(ETH_HLEN)
```

```
backend_addr = ctx_load_meta(CB_ADDR)
dsr_set_ipip4()
host_ip = lookup_ip4_remote_endpoint(backend_addr);
ctx_store_bytes()
revalidate_data();
fib_redirect_v4();
return XDP_TX;
```

HEAD =>
```
outer_ip4->saddr = 10.249.188.39
outer_ip4->daddr = 10.244.65.61
outer_ip4->protocol = IPPROTO_IPIP
inner_ip4->saddr = 10.248.46.20
inner_ip4->daddr = 10.33.4.36
inner_ip4->protocol = IPPROTO_TCP
```

node1
10.244.65.61

XDP

eth0

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_extract_tuple()

+if (tuple->nexthdr == 4)
+    revalidate_data_l3_off(ETH_HLEN + ipv4_hdrlen(ip4));

lb4_lookup_service()
nodeport_svc_lb4()
lb4_local()
lb4_select_backend_id() -> !backend_local
ctx_store_meta(CB_ADDR, backend->address);
CILIUM_CALL_IPV4_NODEPORT_DSR
revalidate_data_l3_off(ETH_HLEN)
```

```
HEAD => outer_ip4->saddr = 10.249.188.39
        outer_ip4->daddr = 10.244.65.61
        outer_ip4->protocol = IPPROTO_IPIP
        inner_ip4->saddr = 10.248.46.20
        inner_ip4->daddr = 10.33.4.36
        inner_ip4->protocol = IPPROTO_TCP
```

```
backend_addr = ctx_load_meta(CB_ADDR)
dsr_set_ipip4()
host_ip = lookup_ip4_remote_endpoint(backend_addr);
ctx_store_bytes()
revalidate_data();
fib_redirect_v4();
return XDP_TX;
```

```
HEAD => outer_ip4->saddr = 10.249.188.39
        outer_ip4->daddr = 10.244.65.61
        outer_ip4->protocol = IPPROTO_IPIP
        inner_ip4->saddr = 10.248.46.20
        inner_ip4->daddr = 10.33.4.36
        inner_ip4->protocol = IPPROTO_TCP
```

node1
10.244.65.61

xDP

eth0

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4()
lb4_extract_tuple()

+if (tuple->nexthdr == 4)
+    revalidate_data_l3_off(ETH_HLEN + ipv4_hdrlen(ip4));

lb4_lookup_service()
nodeport_svc_lb4()
lb4_local()
lb4_select_backend_id() -> !backend_local
ctx_store_meta(CB_ADDR, backend->address);
CILIUM_CALL_IPV4_NODEPORT_DSR
revalidate_data_l3_off(ETH_HLEN)
```

```
HEAD => outer_ip4->saddr = 10.249.188.39
        outer_ip4->daddr = 10.244.65.61
        outer_ip4->protocol = IPPROTO_IPIP
        inner_ip4->saddr = 10.248.46.20
        inner_ip4->daddr = 10.33.4.36
        inner_ip4->protocol = IPPROTO_TCP
```

```
backend_addr = ctx_load_meta(CB_ADDR)
dsr_set_ipip4()
host_ip = lookup_ip4_remote_endpoint(backend_addr);
ctx_store_bytes(saddr);
ctx_store_bytes(daddr);
revalidate_data();
fib_redirect_v4();
return XDP_Tx;
```

```
HEAD => outer_ip4->saddr = 10.244.65.61
        outer_ip4->daddr = 10.244.66.75
        outer_ip4->protocol = IPPROTO_IPIP
        inner_ip4->saddr = 10.248.46.20
        inner_ip4->daddr = 10.33.4.36
        inner_ip4->protocol = IPPROTO_TCP
```

node1
10.244.65.61

	outer	inner
IPIP	10.244.65.61 -> 10.244.66.75	10.248.46.20 -> 10.33.4.36:9090

xDP

eth0

```
<cil_from_netdev>
CILIUM_CALL_IPV4_FROM_NETDEV
nodeport_lb4(l3_off)
lb4_extract_tuple()

+if (tuple->nexthdr == 4)
+    revalidate_data_l3_off(ETH_HLEN + ipv4_hdrlen(ip4));
```

```
HEAD => outer_ip4->saddr = 10.249.188.39
        outer_ip4->daddr = 10.244.65.61
        outer_ip4->protocol = IPPROTO_IPIP
        inner_ip4->saddr = 10.248.46.20
        inner_ip4->daddr = 10.33.4.36
        inner_ip4->protocol = IPPROTO_TCP
```

```
lb4_lookup_service()
nodeport_svc_lb4(l3_off)
lb4_local()
lb4_select_backend_id() -> backend_local
b4_dnat_request(l3_off);
lb4_xlate(l3_off);
ctx_store_bytes(...);
+ctx_adjust_hroom() ?
CTX_ACT_OK;
```

```
HEAD => outer_ip4->saddr = 10.249.188.39
        outer_ip4->daddr = 10.244.65.61
        outer_ip4->protocol = IPPROTO_IPIP
        inner_ip4->saddr = 10.248.46.20
        inner_ip4->daddr = 10.33.4.36
        inner_ip4->protocol = IPPROTO_TCP
```

SKB

lxcXYZ

towards bpf_lxc

backend
10.89.189.131:80

node1
10.244.65.61